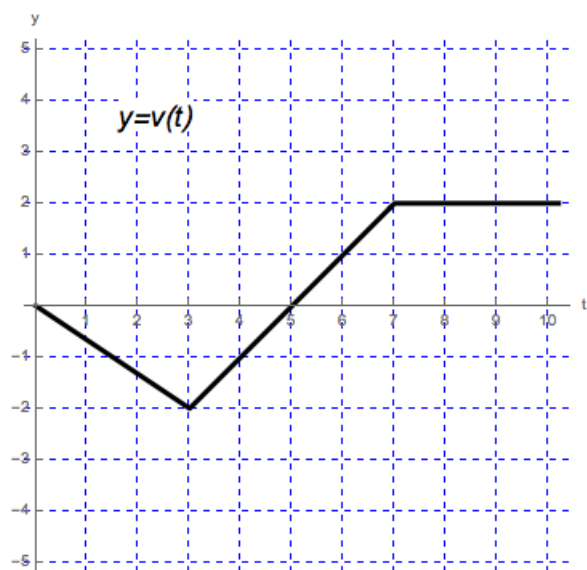


Problem 1

Problem 1

Problem 1 The graph of the velocity (t in min, v in ft/min) of a particle moving along a straight line is given in the figure. Assume that the particle was at the origin initially.



- (a) Evaluate the displacement of the particle over the following intervals.
 - (i) $[0, 10]$
 - (ii) $[0, 7]$
 - (iii) $[0, 5]$
- (b) When was the particle farthest from the origin? $[3]$
- (c) What was the total distance the particle has travelled over the time interval $[0, 10]$?